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Q1

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The key difference between the two is that while list are mutable, tuple are immutable. This means that list can be modified, sliced, appended or constant, but tuple remain modified. & cannot be the

②

Python is provided with us a number of built-in data structures such as list, tuple, set, dictionary, etc. to store & organize the data efficiently.

Q2

⇒

A set is a built-in data type used to store a collection of unique elements. & Set is a type of sequence data type. Following are the purposes of the data type in Python.

- i) Mathematical set operations
- ii) mutable
- iii) Interability
- iv) Hashable element.

2) Example :

Set

$set_1 = \{ \text{"Monday"}, \text{"Tuesday"}, \text{"Wednesday"}, \text{"Thursday"}, \text{"Friday"}, \text{"Saturday"}, \text{"Sunday"} \}$   
 print(set<sub>1</sub>)

// Output: { "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday" }

Q3.

⇒

- Integer numbers are used to represent whole numbers without any decimal points, Float point numbers with the decimal place.

Integer is a numeric data type representing whole numbers without any decimal point.

2)

- Float is a numeric data type representing decimal numbers. Float are used in the calculation of mathematical operations.

Example: i) Integer:  $x = 5$

ii) Float:  $y = 3.14$

Q4  
=>

The python dictionary is built in data type. Dictionaries contain a pair of keys & their corresponding values. The used to store key values where such key is unique within the dictionary.

② Syntax:

Defining using {} brackets with the value separate by a using colon (:).

List: used to sequential data storage where element can be added, removed

Tuple: used to immutable of the tuple.

Q5.

=>

In python a docstring is the special kind of string. It is used to document modules, functions, classes & methods.

They provided a convenient way to associate documentation with python code. Here's a detailed explanation

- i) Introspection
- ii) Documentation Generators
- iii) Help Function

Documentation of reliability is a main purpose of the docstring.

Q6)

⇒ In python, operators are special symbols or keywords used to perform operations on operands.

- ① ✓ Operators are essential in python for performing various types of computation, logical operation.

Q7

⇒

i) The `==` operator is used in python to check if the values of two operands are equal.

It returns to `True` if the values are equal & `False` otherwise.

✓ ii) `is` operator: Compares the identities of two objects. It checks if both variables point to the same object in memory.

Understanding the difference between `==` is crucial for writing correct & efficient python code.

```
ex: a = [1, 2, 3]
    b = [1, 2, 3]
    print(a == b)
```

o/p: True

Q8.

⇒ The `+=` operator is used to accumulate the sum of numbers. The `+=` operator is a convenient & efficient way to update the value of a variable by adding it & its simplifies the code by avoiding the need to repeat the variable name.

Q9

⇒ The `in` operator is used to check for the membership in sequence like string, lists, tuple or dictionary.

Example

```
if world in "hello world":  
    print("Substring found")
```

Q10

⇒ if statement is used to make decision based on conditions. It allow you to execute block of code only if a certain condition is true.

Example :

```
x = 10
```

```
if x > 5
```

```
    print("x is greater than 5")
```

Q12  
⇒

while loop requires a condition that when false, terminate the loop, whereas for loop, iterate over a sequence until the sequence is exhausted. ex

②

✓ while loop requires explicit initialization of variable outside the loop. whereas for loop, iterate over a sequence until the sequence is exhausted.